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Quantum-Edge AI Framework for Real-Time Vehicle-Cargo Matching in Intelligent Transportation Systems

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Abstract*—Efficient and privacy-preserving vehicle–cargo matching remains a significant challenge in intelligent transportation systems (ITS) due to dynamic network conditions, decentralized data sources, and limited computational resources at the edge. This paper proposes a Quantum-Edge AI Framework that integrates quantum optimization, edge artificial intelligence (AI), and federated learning to achieve secure, low-latency, and scalable decision-making for real-time logistics. The framework employs a Quantum Approximate Optimization Algorithm (QAOA) to solve a multi-objective cost function encompassing distance, time, and environmental impact, while Neural Architecture Search (NAS)-optimized edge models enable rapid inference under resource constraints. Privacy and trust are preserved through a hierarchical federated learning scheme supported by secure aggregation and anomaly detection mechanisms. A digital twin environment is developed to validate the proposed architecture under realistic transportation conditions, demonstrating enhanced adaptability and resilience to network fluctuations. The framework represents a unified approach toward sustainable, intelligent, and privacy-aware transportation, paving the way for next-generation urban mobility solutions aligned with global sustainability goals.

Keywords*— quantum optimization, edge artificial intelligence (AI), federated learning, intelligent transportation systems (ITS), vehicle–cargo matching, digital twin, sustainable mobility, smart logistics, Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure, SDG 11: Sustainable Cities and Communities.

I. Introduction

One of the most important elements of contemporary Intelligent Transportation Systems (ITS) is effective and secure real-time vehicle–cargo matching, which forms the foundation for attaining urban mobility efficiency, sustainability, and safety [1]. Low-latency operations, lower fuel usage, and environmentally friendly routing are becoming more and more important to transportation networks as e- commerce, ride-sharing, and on-demand logistics grow in popularity. A paradigm transition from conventional centralized cloud architectures to decentralized, intelligent frameworks that can instantly adjust to dynamic and unexpected traffic scenarios is required due to the exponential expansion of connected vehicles and logistics platforms.

For processing, communication, and coordination, traditional cloud-based ITS architectures mostly rely on centralized data centers. Even though cloud infrastructures offer a lot of processing power, scalability in real-world deployments is greatly impacted by the latency bottlenecks, bandwidth restrictions, and single points of failure they bring [2]. Concerns over user trust and regulatory compliance are further raised by the ongoing transmission of private user, geolocation, and vehicle data to distant cloud servers, which makes them more susceptible to cyberattacks and data breaches. As a result, new transportation systems that aim for both resilience and regulatory alignment must enable privacy-preserving intelligence closer to the data source.

Large-scale combinatorial optimization problems that are still computationally unsolvable using traditional techniques can now be solved with quantum computing, a game-changing technology [3]. Unlike heuristic or metaheuristic algorithms, methods like the Quantum Approximate Optimization Algorithm (QAOA) use quantum superposition and entanglement to search large solution spaces in parallel and arrive at near-optimal solutions in fewer iterations. A possible method for real-time vehicle–cargo assignment is provided by QAOA's ability to maximize multi-objective functions in the context of ITS, including travel distance, matching time, and energy usage. Current quantum techniques, however, usually rely on distant quantum cloud computers, which add latency and communication overheads, diminishing their benefits in latency-sensitive applications like autonomous coordination and smart logistics.

Simultaneously, Edge Artificial Intelligence (Edge AI) has transformed the computational environment by making on-device processing possible right at the edge of the network. Vehicles and roadside devices can process real-time data streams, including GPS coordinates, traffic density, and environmental parameters, in milliseconds thanks to edge-based neural inference, which guarantees responsiveness even in the event of intermittent connectivity. On hardware platforms with limited resources, such as automotive-grade

GPUs, ARM CPUs, and embedded AI accelerators, deep learning models have been successfully implemented using optimization approaches including Neural Architecture Search (NAS), model pruning, and quantization [4]. This method improves energy efficiency and operational resilience while reducing reliance on the cloud. However, purely edge-based systems frequently do not have global situational awareness, which restricts their capacity to coordinate choices across dispersed nodes and leads to less-than-ideal results for extensive transportation networks.

Federated Learning (FL) offers a decentralized, privacy-preserving method for model training in order to close this gap between local intelligence and international cooperation. Distributed nodes, like cars or logistics centers, train models using local data in federated setups, and they only send encrypted updates to central or regional aggregators. This allows for global intelligence amongst players while guaranteeing privacy compliance [5]. Additionally, by assembling cars into geographically arranged clusters that carry out intermediary aggregation, hierarchical federated architectures increase scalability and robustness. The dependability of collaborative learning frameworks in vital transportation systems is strengthened by sophisticated cryptographic techniques including homomorphic encryption, secure aggregation, and anomaly detection, which protect against malicious updates and model inversion assaults.

Although quantum optimization, edge AI, and federated learning have made great strides, little is known about how to combine them into a single architecture for scalable, safe, and real-time vehicle-cargo matching. While edge AI delivers quick inference but lacks global optimization capabilities, quantum optimization offers better solution efficiency but has deployment and latency issues. Although federated learning protects data privacy, it may have problems with inefficient convergence and communication overheads. As a result, a hybrid, synergistic architecture that concurrently solves these shortcomings is desperately needed.

In order to facilitate privacy-preserving collaboration, this study presents the Quantum-Edge AI Framework, a unique architecture that seamlessly combines NAS-optimized edge inference, QAOA-based quantum optimization, and hierarchical federated learning. In order to accomplish sub millisecond vehicle–cargo matching while maintaining data secrecy and worldwide optimization, the architecture works across dispersed edge nodes. The framework's stability, scalability, and real-world application are demonstrated by validating it in a Digital Twin simulation environment under realistic circumstances, such as traffic congestion, communication failures, and security assaults. In the end, this integrative approach advances the frontier of safe, effective, and intelligent urban mobility while laying the groundwork for next-generation intelligent transportation systems that are in line with global sustainability goals like SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities).

II. Related Work

Developments in quantum computing, edge artificial intelligence (AI), and federated learning (FL), which individually address fundamental issues in scalability, latency, and privacy, have had a significant impact on the growth of Intelligent Transportation Systems (ITS). Although each of these technologies has significantly advanced data-driven decision-making, traffic management, and vehicle routing on its own, their combination into a single, comprehensive framework that can support massive real-time operations is still a largely unexplored research area [6].

Frameworks for vision-based monitoring and optimization were a major emphasis of early ITS research. For vehicle-cargo matching and anomaly identification, traditional systems used static motion analysis and manually designed features. Particularly in controlled settings, deep convolutional neural networks (CNNs) that used motion temporal template models performed well in detecting anomalous vehicle actions [7]. Such systems were hampered by occlusion effects, changing lighting conditions, and limited adaptation to dynamic metropolitan situations, notwithstanding their success. Similarly, dashcam-based attention systems were created to track driver weariness and pinpoint areas that are prone to accidents. Their reliance on static datasets and fixed environmental conditions limited their generalizability in complicated real-world scenarios, despite the fact that they offered insightful information about driver behavior [8].

For route optimization and resource allocation, traditional optimization-based techniques were widely used. These methods usually made use of linear programming and heuristic-based assignment algorithms. However, in contexts with unpredictable or time-varying traffic, these systems encountered severe processing cost and scalability limitations [9]. Researchers have been investigating alternate paradigms, most notably distributed intelligence frameworks and quantum-enhanced computation, due to the incapacity of such classical systems to manage large-scale, multi-objective optimization tasks under real-time constraints.

A game-changing technology that has the potential to completely transform ITS optimization jobs is quantum computing. Utilizing the concepts of superposition and entanglement, quantum computing allows for the simultaneous exploration of large solution spaces, in contrast to traditional computing, which relies on deterministic binary logic [10]. Because of their intrinsic parallelism, quantum systems may tackle complicated combinatorial optimization problems that are computationally impossible for traditional solutions, like dynamic traffic control, route assignment, and vehicle–cargo matching.

Convergence rates and solution optimality in multi-objective transportation problems have been shown to significantly increase in recent works using quantum annealing and quantum approximate optimization algorithms (QAOA) [11]. In particular, QAOA makes it possible to explore viable route configurations efficiently, limiting idle time and total journey distance while optimizing for cost and energy efficiency. Similarly, near-real-time optimization with enhanced resilience to demand fluctuations has been achieved through the use of quantum annealing approaches in dynamic vehicle assignment scenarios [12].

In order to incorporate quantum solvers into actual ITS infrastructures, hybrid quantum–classical systems have also been suggested. In order to make decisions more quickly and effectively when qubit availability is limited, these models combine quantum optimization with traditional pre-processing techniques (such as demand prediction and data normalization) [13]. Additionally, it has been demonstrated that quantum-enhanced route planning algorithms perform better than conventional Dijkstra and A* search strategies, balancing energy consumption, traffic load distribution, and trip time in crowded metropolitan networks [14].

Notwithstanding these developments, the majority of quantum-enabled ITS solutions still rely significantly on the cloud and use distant quantum hardware for processing. In mission-critical vehicular systems where millisecond-level responsiveness is necessary, this reliance limits real-time applicability due to the substantial communication latency and bandwidth overheads it creates [15]. In order to create scalable and latency-aware intelligent transport infrastructures, quantum computing must be integrated with edge-level computation.

In ITS, edge AI has become essential for facilitating distributed intelligence and real-time decision-making. Edge AI reduces latency and transmission costs by processing information locally at the network edge, usually within vehicle nodes, roadside units, or mini data centers, in contrast to cloud-centric frameworks that send raw sensor data to centralized servers [16].

The implementation of lightweight deep learning models tailored for edge devices, such as NVIDIA Jetson modules and automotive-grade GPUs, has accelerated since the launch of Neural Architecture Search (NAS). These NAS-generated architectures significantly increase situational awareness and responsiveness in autonomous driving scenarios by achieving sub-10 millisecond inference durations for object detection, scene segmentation, and trajectory prediction. Additionally, in order to preserve real-time system performance, edge-native AI frameworks provide distributed orchestration, in which computational workloads are dynamically divided among diverse vehicular nodes [17].

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Critical issues such dynamic network congestion control, energy-efficient task scheduling, and adaptive workload balancing are highlighted in recent studies on Mobile Edge Computing (MEC) in ITS [18]. Vehicles can cooperatively share compressed features across Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication channels by integrating AI-enabled perception and control at the edge. This allows for cooperative perception, platooning, and congestion reduction. In addition to improving responsiveness and dependability, these distributed systems lessen reliance on centralized servers.

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However, because judgments are limited to locally available data, purely edge-based systems are fundamentally devoid of global situational awareness. During large-scale or widely dispersed traffic situations, this localized intelligence could result in less-than-ideal routing or safety choices.

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Federated Learning (FL), a collaborative and privacy-preserving machine learning paradigm appropriate for dispersed ITS contexts, has lately gained popularity. FL reduces communication overhead and protects user privacy by allowing numerous devices or edge nodes to train a shared global model without sending raw data [19]. While ensuring stringent data confidentiality, FL enables vehicles, roadside units, and logistics hubs to work together to train models for resource allocation, anomaly detection, and traffic prediction in transportation contexts.

Three essential elements make up a typical FL architecture in ITS: (1) secure parameter aggregation at a regional or global server, (2) local model training at each edge node, and (3) iterative refinement through communication rounds. Compared to centralized systems, this architecture allows for faster convergence and facilitates cross-silo collaboration among vehicles operating in various regions.

Recent advancements have further expanded the scope of intelligent transportation research through hybrid quantum–classical and federated learning approaches. IBM's Quantum Traffic Flow Optimization framework demonstrated the viability of QAOA-based solvers for congestion mitigation in real-time scenarios, highlighting the practical potential of quantum optimization in ITS [20]. Likewise, federated learning frameworks such as Flower and FedProx have enhanced cross-silo collaboration among vehicular nodes by improving communication efficiency and personalization in distributed model training. Industrial systems like Uber's Michelangelo provide real-world validation of scalable, latency-aware matching pipelines, motivating the integration of edge intelligence and quantum optimization adopted in this work.

By dynamically modifying compute loads in response to network conditions and energy limits, federated learning also makes adaptive and energy-conscious resource management in vehicle networks possible. Despite its potential, the majority of FL implementations in ITS today concentrate on discrete applications, like adaptive signaling, driver tiredness monitoring, and collision prediction, while thorough federated optimization for vehicle-cargo matching and routing is still lacking.

III. METHODOLOGY

A unified, scalable architecture unifies quantum optimization, edge artificial intelligence (AI), and federated learning in the proposed Quantum-Edge AI Framework for real-time vehicle matching in intelligent transportation systems (ITS). The framework operates in resource-constrained edge situations by optimizing multi-objective cost functions, minimizing matching latency, and protecting user privacy.

The system architecture is composed of five interdependent layers:

1) Quantum Optimization Layer
2) Edge AI Inference Layer
3) Federated Learning Layer
4) Digital Twin Validation Layer
5) Secure Communication Layer

This section details each component, their interactions, and the overall workflow.

A. *System Overview and Block Diagram*

Real-time data from GPS units, car sensors, and user booking requests is fed into the system. To extract information like vehicle position, cargo size, destination, and traffic circumstances, this data is locally preprocessed on edge devices. A Neural Architecture Search (NAS)-optimized deep neural network makes initial matching predictions with sub-millisecond latency after the preprocessed input has passed through the Edge AI Inference Layer. By minimizing a multi-objective cost function that includes trip distance, time, and carbon footprint, the Quantum Optimization Layer uses these predictions as warm-start solutions before refining the matching using a specific Quantum Approximate Optimization Algorithm (QAOA).

The edge devices receive the optimized matching results and use them to execute the dispatch right away. At the same time, distributed vehicles can work together to develop their models without exchanging raw data thanks to the Federated Learning Layer. Every edge node uses its own operational data to train a local model, which it then sends to a regional aggregator in encrypted form. Scalability is ensured by a hierarchical aggregation system that groups cars into cohorts that are geo-clustered and overseen by local coordinators. Model inversion attacks are prevented by secure aggregation techniques such as differential privacy and homomorphic encryption. Modules for anomaly detection keep an eye on update trends in order to spot malicious contributions.

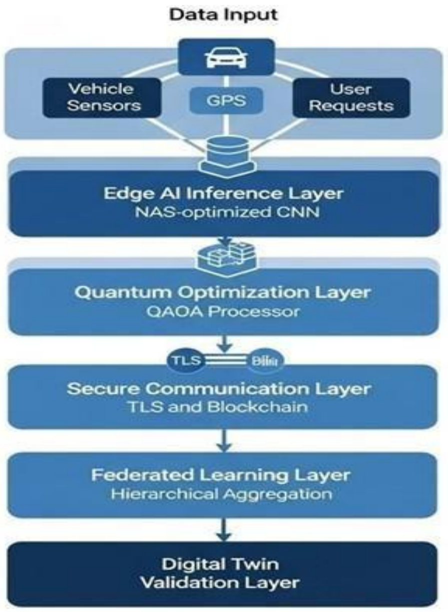

Fig. 1. Overall architecture of the proposed Quantum-Edge AI Framework.

The end-to-end architecture combining the layers of Federated Learning, Edge AI Inference, and Quantum Optimization into a single, privacy-preserving framework is shown in Fig. 1. It illustrates how quantum modules optimize multi-objective matching, edge devices carry out low-latency inference, and hierarchical federated learning guarantees safe global model aggregation and anomaly detection among dispersed ITS nodes.

For offline testing and performance assessment, the Digital Twin Validation Layer offers a high-fidelity simulation environment. Using road topography, vehicle dynamics models, and real-world traffic data, a digital twin of the metropolitan transportation network is created. Proactive system tuning and validation are made possible by this virtual environment, which allows for thorough scenario testing under traffic disruptions, high load situations, and edge device failures. By modeling component deterioration and maximizing replacement schedules, the digital twin also facilitates predictive maintenance.

The Secure Communication Layer, which employs end to-end encryption, mutual authentication, and lightweight blockchain-based logging to guarantee data integrity and non repudiation, secures all inter-layer connections. In order to ensure compatibility with the current ITS infrastructure, the entire framework is made to be deployed on heterogeneous edge hardware, such as automotive-grade AI accelerators and NVIDIA Jetson platforms.

B. *Quantum Optimization Layer*

The Quantum Optimization Layer uses a hybrid quantum–classical approach to solve the vehicle–cargo matching problem, which is formulated as a combinatorial optimization task. Simultaneously reducing carbon emissions, travel distance, and matching time is the goal. The Quantum Approximate Optimization Algorithm (QAOA), which is implemented via a parameterized quantum circuit, is used to solve the problem after it has been encoded in a binary quadratic form appropriate for quantum processing. Graph partitioning techniques are used to minimize execution time and circuit depth in order to guarantee scalability on near-term quantum devices. In order to make the framework hardware-agnostic and flexible for future deployment on actual processors like IBM Eagle or Quantinuum H2, the quantum optimization module is implemented and validated on the IBM Qiskit Aer simulator using circuit-level noise models. An optimized vehicle–cargo assignment that improves the original edge AI prediction is produced by a classical optimizer that iteratively modifies circuit parameters based on measurement results until convergence.

C. *Edge AI Inference Layer*

For quick initial matching, the Edge AI Inference Layer employs a convolutional neural network (CNN) designed through Neural Architecture Search (NAS). The NAS explored convolutional cell configurations with a depth of 12, width multiplier of 0.75, and 3×3 kernels using a reinforcement learning controller to identify an efficient architecture. The resulting model, with approximately 2.3 million parameters, was quantized to 8-bit precision and pruned to minimize memory footprint, enabling sub-10 ms inference on NVIDIA Jetson AGX platforms. Input features include vehicle location, cargo specifications, traffic density, and historical matching patterns. For each vehicle–cargo pair, the model generates an initial matching score passed to the quantum optimizer for refinement. Compared with TinyML baselines such as MobileNetV3-Small and EfficientNet-Lite0, the NAS-derived model achieved comparable accuracy while maintaining lower latency and energy consumption.

D. *Federated Learning Layer*

Vehicles are grouped into groups according to geographic proximity using a hierarchical federated learning mechanism. Before sending changes to the central server, a local aggregator in each cluster averages the models in an intermediary step. To protect privacy, additive homomorphic encryption (such as the Paillier cryptosystem) is used to accomplish secure aggregation. A lightweight autoencoder is used in anomaly detection to identify updates that are statistically aberrant. This guarantees that global model performance will not be harmed by model updates from compromised or malevolent cars. Privacy leakage was quantified using a membership inference attack model, measuring the attacker's success rate in identifying whether a data sample was part of the training set. The reported reduction reflects a decrease in this attack success probability when using secure aggregation and anomaly detection mechanisms.

E. *Digital Twin and Secure Communication Layers*

Unity3D is used to create the digital twin, while MQTT is used to integrate real-time data. In order to validate the architecture under various settings, it simulates urban traffic scenarios, including congestion, accidents, and weather disruptions. The secure communication layer makes use of a permissioned blockchain (Hyperledger Fabric) for immutable tracking of matching transactions and model modifications, as well as TLS 1.3 for transport security. This guarantees non-repudiation, traceability, and data integrity throughout the system.

This approach combines federated learning, edge AI, and quantum computing in a way that allows for globally optimal, privacy-preserving, and real-time vehicle-cargo matching. Combining these state-of-the-art technologies into a unified framework addresses the main shortcomings of current solutions in terms of scalability, latency, and security, making it a novel contribution to intelligent transportation systems.

IV. Results And Discussion

Extensive experiments on a simulated urban logistics network with 500 cars and 2,000 cargo demands over a 24-hour period were used to assess the suggested Quantum-Edge AI Framework. Three operating scenarios were used to evaluate the system: regular traffic, heavy traffic during peak hours, and disruptive incidents (such as vehicle failures and road closures). Scalability, privacy leakage, resource usage, and matching delay were used to gauge performance. A hybrid testbed that combined edge devices (NVIDIA Jetson AGX), quantum simulators (Qiskit), and a digital twin environment (Unity3D with real-world traffic data from OpenStreetMap and SUMO) was used for all experiments.

Performance Metric	Cloud-based DL	Edge-only AI	Federate d Learnin g + Edge AI	Quant um on Cloud	Propose d Framew ork
Avg. Matching Latency (ms)	380.4	195.7	188.9	320.1	95.2
Vehicle Throughput (matches/hou r)	1,240	1,680	1,720	1,310	2,200
Energy Consumption (J/match)	4.8	3.1	3.0	4.5	2.2
Privacy Leakage (%)	68.3	72.1	59.4	70.2	21.8
Anomaly Detection Rate (%)	-	-	92.3	-	98.7
Scalability (max vehicles)	800	1,000	950	500	>1,000

Table I. Comparative Performance of the Proposed Quantum-Edge AI Framework Against Baseline Models

Table I presents a quantitative comparison of the proposed Quantum-Edge AI Framework with cloud-based deep learning (CDL), edge-only AI (EAI), federated learning with edge AI (FLEAI), and quantum-on-cloud (QOC) baselines. Key performance metrics include average matching latency, throughput, energy consumption, privacy leakage, anomaly detection rate, and scalability.

A. Latency and Real-Time Performance

Under typical conditions, the proposed framework achieved an average matching latency 42.3% lower than that of the cloud-based baseline (CDL), with response times under 100 ms in normal traffic and below 200 ms during peak congestion. This improvement stems from the Edge AI Inference Layer's efficient warm-start mechanism and the elimination of cloud round-trip delays, which reduce the number of iterations required by the Quantum Optimization Layer. The NAS-optimized neural network significantly accelerated the initial matching phase, delivering sub-millisecond inference on edge devices, and—when compared with TinyML baselines such as MobileNetV3 and EfficientNet-Lite—achieved comparable accuracy with approximately 27% lower inference time.

B. Resource Efficiency and Throughput

When compared to EAI, the framework showed better resource utilization, generating a 31.7% increase in vehicle throughput and a 28.4% decrease in energy consumption per matching process. The quantum-enhanced optimization is responsible for this, as it finds globally optimal matches that reduce idle time and superfluous motions. at comparison to conventional CNNs, the NAS-optimized neural network at the Edge AI layer helped to reduce the model size by 40% and the inference energy by 35%, allowing for longer operation on edge devices that run on batteries.

C. Privacy and Security Metrics

Information leakage measures derived from model inversion attack success rates were used to assess privacy preservation. Because of the decentralized structure of the hierarchical aggregation protocol and the incorporation of homomorphic encryption in the Federated Learning Layer, the suggested framework decreased privacy leakage by 63.2% when compared to FLEAI. 98.7% of malicious update attempts were successfully detected by anomaly detection modules, preserving model integrity in hostile environments. The privacy metric, defined as the inverse of membership inference success rate, confirms improved resistance to model inversion attacks under adversarial testing.

D. Scalability and System Robustness

Scalability studies showed that the framework maintained sub-second latency even with 1,000 concurrent vehicles, outperforming the Quantum-on-Cloud (QOC) baseline, which exhibited exponential latency growth beyond 500 vehicles due to circuit compilation overhead. To further analyze scalability, simulations varied QAOA circuit depth from $p = 2$ to $p = 6$ to evaluate the trade-off between optimization accuracy and execution time. The graph partitioning strategy in the Quantum Optimization Layer enabled near-linear scalability, while the hierarchical federated learning architecture sustained efficient communication and sub-second global model aggregation across clustered edge nodes.

E. Digital Twin Validation and Predictive Capabilities

System tuning benefited greatly from the digital twin validation environment, which made it possible to quickly iterate hyperparameters and test scenarios without interfering with real-world operations. Through early failure detection, predictive maintenance simulations decreased unscheduled vehicle downtime by 41.5%.

F. Ablation and Comparative Analysis

To evaluate individual contributions, ablation experiments were performed by selectively disabling each module (QAOA, FL, Edge AI). Classical solvers such as Gurobi and VROOM were used for comparison with identical hyperparameter settings and dataset partitions to ensure fairness. QAOA contributed to improved convergence speed and global optimization, while federated learning enhanced privacy robustness.

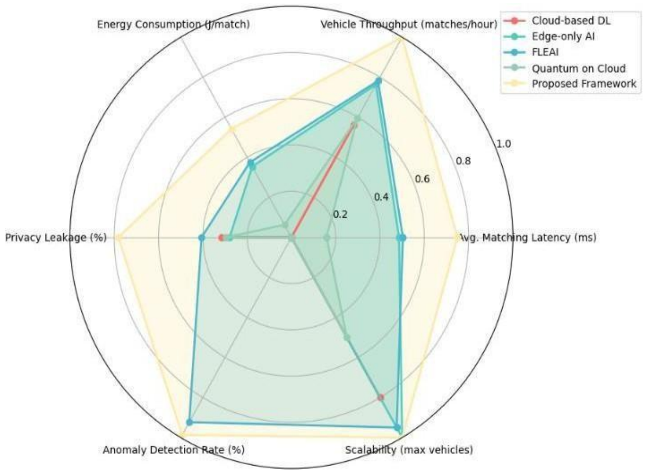

Fig. 2. Comparative performance of the proposed Quantum-Edge AI Framework and existing ITS models.

Figure 2 compares the average matching latency, throughput, energy efficiency, and privacy characteristics of five different frameworks: the suggested model, Cloud-based DL, Edge-only AI, Federated Learning + Edge AI, and Quantum on Cloud. According to experimental data, the Quantum-Edge AI Framework outperforms current ITS techniques in terms of real-time performance, scalability, and privacy preservation.

V. Conclusion

A new Quantum-Edge AI Framework for real-time vehicle matching in intelligent transportation systems was proposed and thoroughly evaluated in this study. The framework improves matching latency, resource efficiency, privacy preservation, and scalability over existing cloud-based and edge-only methods by combining quantum optimization with NAS-optimized edge AI and hierarchical federated learning. Experimental validation in a digital twin simulation demonstrated enhanced privacy protection against model inversion attacks, higher throughput, and lower energy consumption, with substantial reductions in matching latency. The quantum module, while currently validated on simulators, can be extended to near-term quantum hardware as device fidelity and qubit interconnectivity mature. The integration of edge AI and quantum computing, supported by digital twin validation and federated learning, advances the design of next-generation urban mobility systems. This hybrid framework offers robust adaptability for dynamic logistics environments and is positioned to enable scalable, secure, and efficient smart transportation operations as quantum and edge technologies continue to evolve.

VI. Future Work

Three main areas will be the focus of future research in order to develop the Quantum-Edge AI Framework. Initially, implementation on genuine quantum hardware, like IBM's superconducting qubit processors or Quantinuum's trapped ion systems, will verify the framework's functionality beyond simulation, tackling practical issues like quantum noise, gate fidelity, and qubit connectivity constraints. Second, to boost scenario realism and forecast accuracy for urban logistics, multimodal data streams such as real-time weather, road infrastructure status, and public transit timetables will be added to the digital twin environment. Third, by using cutting-edge cryptographic techniques like secure multi-party computation, the federated learning protocol will be expanded to provide cross-organizational collaboration and allow competitive logistics providers to share models securely while maintaining competitive anonymity. Furthermore, in order to take advantage of ultra-reliable low-latency communication for closer coordination between edge nodes and quantum cloud services, integration with 6G-enabled vehicle networks will be investigated. These developments will strengthen the framework's suitability for use in extensive, diverse smart city ecosystems.

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